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(JICSAP/JSA)

**Specification of implementation for
integrated circuit(s) cards—Part 4:
High speed proximity cards**

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Foreword

This translation has been made based on the original Japanese Industrial Standard revised by the Minister of Economy, Trade and Industry, through deliberations at the Japanese Industrial Standards Committee as the result of proposal for revision of Japanese Industrial Standard submitted by Japan IC Card System Application Council (JICSAP)/Japanese Standards Association (JSA) with the draft being attached, based on the provision of Article 12 Clause 1 of the Industrial Standardization Law applicable to the case of revision by the provision of Article 14.

Consequently **JIS X 6319-4:2005** is replaced with this Standard.

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Attention is drawn to the possibility that some parts of this Standard may conflict with a patent right, application for a patent after opening to the public, utility model right or application for registration of utility model after opening to the public. The relevant Minister and the Japanese Industrial Standards Committee are not responsible for identifying the patent right, application for a patent after opening to the public, utility model right or application for registration of utility model after opening to the public.

JIS X 6319 series consists of the following four parts under the general title “*Specification of implementation for integrated circuit(s) cards*”:

Part 1: Integrated circuit cards with contacts

Part 2: Proximity cards

Part 3: Common commands for interchange

Part 4: High speed proximity cards

Specification of implementation for integrated circuit(s) cards— Part 4: High speed proximity cards

Introduction

This Japanese Industrial Standard has been prepared to specify the high speed contactless proximity integrated circuit(s) cards and the proximity coupling devices.

In order to provide the further details for interoperability of contactless interfaces, this Standard provides implementation specifications. Use of new standards in building and operating integrated circuit(s) card systems is in no way constricted because contactless proximity integrated circuit(s) cards have the potential to be used for a variety of applications and systems, while the performance of IC chips, such as power consumption, is constantly being improved by technological advances. Consideration is also given to the possibility that integrated circuit(s) cards with or without contacts may be integrated into the same system.

No International Standard corresponding to this Standard has been established at this point.

1 Scope

This Standard specifies the physical characteristics, air interface, transmission protocols, file structure, commands and the like of high-speed contactless proximity integrated circuit cards (hereafter referred to as “PICCs”).

The PICCs offer the functionality for processing multiple files at one time and handling power disruption during processing with reduced transactions, while maintaining the security between those files.

The PICCs are used in systems such as electronic ticketing at railway stations and event sites, which require high-speed and processing of accesses with quickly moving users are essential.

2 Normative references

The following standards contain provisions which, through reference in this text, constitute provisions of this Standard. The most recent editions of the standards (including amendments) listed below shall be applied.

JIS X 5211 *Information technology—Telecommunications and information exchange between systems—Near Field Communication—Interface and Protocol(NFCIP-1)*

JIS X 6301 *Identification cards—Physical characteristics*

JIS X 6305-1 *Identification cards—Test methods—Part 1: General characteristics*

JIS X 6305-6 *Identification cards—Test methods—Part 6: Proximity cards*

JIS X 6320-4 *Identification cards—Integrated circuit cards—Part 4: Organization, security and commands for interchange*

JIS X 6322-1 *Identification cards—Contactless integrated circuit(s) cards—Proximity cards—Part 1: Physical characteristics*

JIS X 6322-2 *Identification cards—Contactless integrated circuit(s) cards—Proximity cards—Part 2: Radio frequency power and signal interface*

JIS X 6322-3 *Identification cards—Contactless integrated circuit(s) cards—Proximity cards—Part 3: Initialization and anticollision*

JIS X 6322-4 *Identification cards—Contactless integrated circuit(s) cards—Proximity cards—Part 4: Transmission protocol*

3 Terms and definitions, and abbreviations and symbols

For the purposes of this Standard, the terms and definitions, and abbreviations and symbols given in **JIS X 6322-1** to **JIS X 6322-4**, **JIS X 6320-4** and **JIS X 6305-6**, together with the following apply.

3.1 Terms and definitions

3.1.1 access control process category

information on whether pre-authentication is required or not at the time of accessing, and the method for accessing to blocks in service files

3.1.2 area key

key to authenticate the right of access to area files

3.1.3 area file

file to manage the file hierarchy in the PICC

3.1.4 area file structure

block to define the range of the file identifiers of available files, the number of available blocks, and the right of access for the PICC service operator

3.1.5 area file identifier

identifier consisting of the area number and the access control process category

3.1.6 area number

unique number to identify the area file in the PICC

3.1.7 response time descriptor

information to determine the response time for each command

NOTE : In **JIS X 6319-4:2005**, the “response time descriptor” was referred to as the “Pad”.

3.1.8 decrement access

access control process category that subtracts the data in the PICC by writing process in add-subtract access

3.1.9 service key

key to authenticate the right of access to service files

3.1.10 service file

accessing unit when the PCD accesses the PICC, which is the area defined so as to have the same access control process category in the PICC's memory area

3.1.11 service file structure

block to define the access control process category of the relevant service, and the position, range, etc. of the accessible block

3.1.12 service file identifier

identifier consisting of the service number and the access control process category

3.1.13 service number

number to identify the unique service file in the PICC

3.1.14 system code

parameter used for narrowing down the possible PICCs during the communication initialisation

3.1.15 higher layer protocol

protocol layer (not specified in this Standard) to transmit the information which belongs to the protocol (the application or the higher layer of protocol) not defined in this Standard by using the protocol layer specified in this Standard

3.1.16 status flag

flag to indicate the result of processing

The status flag 1 indicates the result of processing or the position where an error has occurred, and the status flag 2 indicates detailed information thereof.

3.1.17 mutual authentication

confirmation of mutual credibility between the PCD and the PICC

3.1.18 time slot

time frame defined for the anticollision among multiple response messages from PICCs in the communication initialisation

3.1.19 direct access

access control process category in which data are overwritten in the PICC by writing process in add-subtract access

3.1.20 power load

load which makes the reference PICC or the PICC under testing consume the energy of operating field

3.1.21 issue ID (IDi)

information written when the PICC is issued

3.1.22 issue parameter (PMi)

additional information written when the PICC is issued

3.1.23 PICC identifier

unique number for the PICC designated by the manufacturer

NOTE: In **JIS X 6319-4:2005**, it was referred to as “NFCID2”.

3.1.24 contactless interface

interface for signal communication or power supply to the PICC without the conductive contact (i.e., without contacting the ICs embedded in the PICC to the PCD)

3.1.25 file identifier

information used for the file identification in the PICC's memory area

The area identifier for area files and the service identifier for service files are included.

3.1.26 block

minimum unit for reading, writing and deleting information in the PICC's memory area

3.1.27 block number

number for management of the logical location of the block which belongs the specific service file

This number shall be used when accessing the memory area of the PICC from the PCD

3.1.28 block list

list indicating the access mode to the block, the block number and the order of service files designated by the list of file identifiers

3.1.29 cash back access

access control process category that allows the data in the PICC to increment within the range of the value decremented in the last access by the writing process in the add-subtract access

3.1.30 request code

parameter used by the PCD to request the additional information from the PICC during the communication initialisation

3.1.31 request data

additional information sent out by the PICC according to the request from the PCD during the communication initialisation

3.2 Abbreviations and symbols

ASK amplitude shift keying

CRC cyclic redundancy check code

EDC error detection code

f_c frequency of operating field (carrier frequency)

H_{max} maximum field strength of the PCD antenna field

H_{min} minimum field strength of the PCD antenna field